Six points with a contractible Rips shadow and nontrivial fundamental group

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Abstract

The affine projection from a Euclidean Vietoris–Rips complex to its shadow need not be injective on fundamental groups. We give six rational points in three-dimensional space whose Rips complex retracts onto a four-cycle, while its shadow is the union of two convex quadrilaterals containing the origin and is therefore contractible. Scaling and isometric inclusion give the same failure in every ambient dimension at least three and at every positive scale. This refutes the injectivity conjecture of Adamaszek, Frick and Vakili.

For a nonempty finite set $X \subset \mathbb{R}^n$ and a scale $r > 0$, let

$$K = \mathrm{VR}_<(X; r) = \{\sigma \subseteq X : \operatorname{diam}(\sigma) < r\}, \qquad S(X; r) = \bigcup_{\sigma \in K} \operatorname{conv}(\sigma),$$

where distances are Euclidean. The vertex inclusion extends affinely on each abstract simplex to a continuous map $p : |K| \to S(X; r)$, called the shadow map. The complex is determined by its edges: a set of vertices spans a simplex exactly when every pair is at distance less than r.

Chambers, de Silva, Erickson and Ghrist proved that the shadow map induces an isomorphism on fundamental groups for planar samples [2, Theorem 3.1]. The strict and non-strict scale conventions are equivalent for these finite-sample statements by choosing a scale in a gap between the finitely many distances [1, Remark 1.4]. Adamaszek, Frick and Vakili proved surjectivity in three-dimensional space [1, Theorem I] and conjectured injectivity in all dimensions [1, Conjecture 7.2]. The following example answers that fixed-sample question negatively already in dimension three.

Theorem 1. *For every integer $n \geq 3$ and every $r > 0$, there is a six-point set $X \subset \mathbb{R}^n$ such that $S(X; r)$ is contractible and*

$$\pi_1(|\,\mathrm{VR}_<(X; r)|, v) \cong \mathbb{Z} \qquad \textit{for every } v \in X.$$

Consequently the affine shadow map is not injective on fundamental groups at any sample base vertex.

Proof. We first take $n = 3$ and $r = 1$. Put

$$a = \frac{9}{20}, \qquad b = \frac{501}{1000}, \qquad s = \frac{309}{1000},$$

and let $X = \{v_0, \ldots, v_5\}$, where

$$v_0 = (a, s, 0), \quad v_1 = (a, -s, 0), \quad v_2 = (-b, 0, 0),$$
$$v_3 = (b, 0, 0), \quad v_4 = (-a, 0, s), \quad v_5 = (-a, 0, -s).$$

All six points are distinct. The following table lists every unordered pair; the notation ij denotes the pair $\{v_i, v_j\}$.

Pairs	Squared distance	Comparison with 1
$01, 45$	$4s^2 = 95481/250000$	< 1
$02, 12, 34, 35$	$(a+b)^2 + s^2 = 499941/500000$	< 1
$03, 13, 24, 25$	$(b-a)^2 + s^2 = 49041/500000$	< 1
$04, 05, 14, 15$	$4a^2 + 2s^2 = 500481/500000$	> 1
23	$4b^2 = 251001/250000$	> 1

In particular, no distance equals the threshold. The only maximal simplices of $K = \mathrm{VR}_<(X; 1)$ are

$$[012], \quad [013], \quad [245], \quad [345]. \tag{1}$$

Indeed, there is no edge between $\{0, 1\}$ and $\{4, 5\}$, and the edge 23 is missing. On each of the vertex sets $\{0, 1, 2, 3\}$ and $\{2, 3, 4, 5\}$, this leaves precisely the two triangles displayed in (1).

Let $C \subset K$ be the four-cycle with ordered vertices $2, 0, 3, 4$. Define a map on abstract vertices by

$$f(1) = 0, \qquad f(5) = 4, \qquad f(i) = i \quad (i \in \{0, 2, 3, 4\}).$$

It sends the four triangles in (1) to the edges $[02], [03], [24], [34]$, respectively, and extends to a simplicial retraction $f : |K| \to |C|$. Moreover, for every simplex σ in (1), the vertices of $\sigma \cup f(\sigma)$ are still contained in σ. In the standard barycentric realization, therefore,

$$H(u, t) = (1 - t)u + tf(u), \qquad 0 \le t \le 1,$$

stays in that same simplex. These affine formulas agree on shared faces, and H fixes $|C|$ pointwise. Thus $|K|$ strongly deformation retracts onto the four-cycle. Its fundamental group is infinite cyclic [3, Theorem 1.7 and Proposition 1.17]; a generator at v_2 is the edge loop

$$\gamma = (v_2, v_0, v_3, v_4, v_2). \tag{2}$$

The shadow has a different topology. Set

$$Q_1 = \mathrm{conv}\{v_0, v_1, v_2\} \cup \mathrm{conv}\{v_0, v_1, v_3\}, \qquad Q_2 = \mathrm{conv}\{v_2, v_4, v_5\} \cup \mathrm{conv}\{v_3, v_4, v_5\}.$$

By (1), the whole shadow is $Q_1 \cup Q_2$. In the xy-plane the first triangle has slices $-b \le x \le a$, $|y| \le s(x + b)/(a + b)$, and the second has slices $a \le x \le b$, $|y| \le s(b - x)/(b - a)$. Since $0 < a < b$, their union is exactly

$$Q_1 = \left\{ (x, y, 0) : |y| \le \frac{s(x + b)}{a + b}, \quad |y| \le \frac{s(b - x)}{b - a} \right\}. \tag{3}$$

The two bounds agree at $x = a$; the first is smaller for $x \le a$ and the second for $x \ge a$. Their nonnegativity forces $-b \le x \le b$. The four linear inequalities in (3) show that Q_1 is a convex quadrilateral. It contains the entire axis segment $[v_2, v_3]$. The linear isometry $(x, y, z) \mapsto (-x, z, y)$ sends Q_1 to Q_2, so Q_2 is also convex and contains that segment.

Both quadrilaterals contain the origin. Hence

$$h(w, t) = (1 - t)w, \qquad w \in Q_1 \cup Q_2, \quad 0 \le t \le 1,$$

is a contraction of the shadow: a point stays in whichever convex quadrilateral originally contains it. The projected loop $p(\gamma)$ is therefore null-homotopic, whereas (2) is nontrivial in $|K|$. The complex is connected, so its fundamental group is infinite cyclic at every sample base vertex [3, Proposition 1.5]; the contractible shadow has trivial fundamental group at all of them.

For $n \geq 3$, append $n-3$ zero coordinates. For any prescribed $r > 0$, multiply the six points by r. Squared distances and the squared threshold are both multiplied by r^2, giving exactly the same abstract complex, while the shadow is carried to the new one by an isometric inclusion and a positive scaling. This proves the theorem. □

The segment $[v_2, v_3]$ explains the lost loop. Its endpoints are too far apart to be a Rips edge, but the segment lies in both planar parts of the shadow. The two abstract triangulated disks meet only in the vertices v_2, v_3; their shadows meet along the whole segment.

Exact verification. The accompanying standard-library Python code enumerates the exact rational distances and all simplices, checks the carried vertex retraction, and checks an integer simplicial cocycle taking value 1 on (2). These finite checks supplement the explicit homotopies above.

References

[1] Michał Adamaszek, Florian Frick and Adrien Vakili, *On homotopy types of Euclidean Rips complexes*, arXiv:1602.04131v1, 2016.

[2] Erin W. Chambers, Vin de Silva, Jeff Erickson and Robert Ghrist, *Rips Complexes of Planar Point Sets*, arXiv:0712.0395v1, 2007.

[3] Allen Hatcher, *Algebraic Topology*, Cambridge University Press, 2002. Original monograph PDF.

AI-use statement. OpenAI Codex was used to develop and analyze the construction, check exact arithmetic, and draft the manuscript and verification code. The mathematical argument is given in full above. Independent mathematical review is requested.